Analytical Chemistry Cumulative Exam
March 2, 2012

This cume is on the following article: Michael J. Moehlenbrock, Robert L. Arechederra, Kyle H. Sjöholm, and Shelley D. Minteer, "Analytical Techniques for Characterizing Enzymatic Biofuel Cells" *Anal. Chem.* **2009**, *81*, 9538–9545.

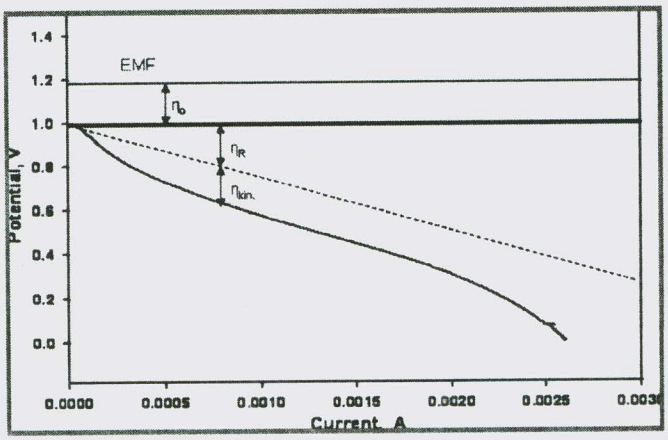
1. (15 points) Describe what an enzymatic biofuel cell is and how it works. Give at least one example of each of the electrochemical cell components: electrodes, enzymes, fuels, and oxidants. What is the motivation behind developing enzymatic biofuel cells? What do people think that the main niche for biofuel cells will be?

2. (12 points) List two spectroscopic methods mentioned in the article for characterizing enzymatic biofuel cells. What information can they provide about the biofuel cell system?

3. (12 points) What are two electrochemical techniques mentioned in the article that are used to characterize enzymatic biofuel cells? What information can they provide about the biofuel cell system?

4. (15 points) Define DET and MET. Compare and contrast these two processes.

5. (15 points) Below is a figure from the article. What is the name of this curve and what does it illustrate? Explain the features of this figure.



6. (14 points) In the article, it is said that, "Many oxidoreductase enzymes require mediators or electrocatalysts to facilitate the transfer of electrons from their co-enzymes [typically NAD(P)+ or FAD to the electrode surface." Name two such mediators or electrocatalysts and their advantages and disadvantages.

7. (12 points) Name two materials characterization methods that are mentioned in the article. Describe in general how they work (what they actually detect), and why they are of interest (their usefulness) in the biofuel cell community.

8. (5 points) Name one analytical application of the biofuel cells, themselves.